

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

Frequency Analysis Attacks

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What Is Frequency Analysis?

- Frequency analysis is a codebreaking technique that examines how often letters appear in a piece of encrypted text.
- In any language, certain letters and words appear more often than other words.
- Frequency analysis was a commonly used codebreaking technique in the times of the Caesar and Vigenere ciphers

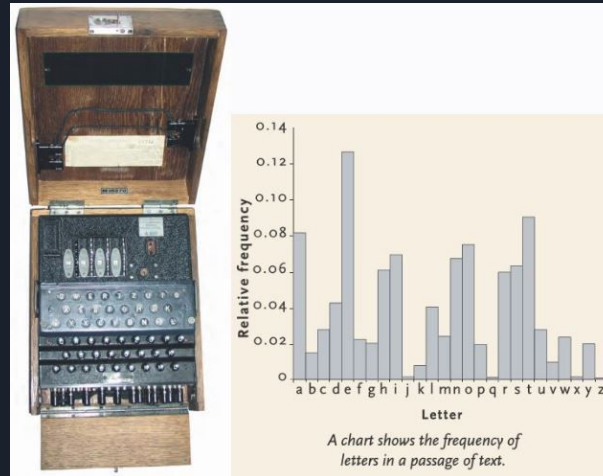
Letters								
e	t	a	o	i	n	s	r	h
12.49%	9.28%	8.04%	7.64%	7.57%	7.23%	6.51%	6.28%	5.05%

Bigrams											
th	he	in	er	an	re	on	at	en	nd	ti	es
3.56%	3.08%	2.43%	2.05%	1.99%	1.85%	1.78%	1.49%	1.45%	1.35%	1.34%	1.34%

Double Letters										
ll	ss	ee	oo	tt	ff	pp	rr	mm	cc	nn
0.58%	0.41%	0.35%	0.21%	0.17%	0.15%	0.14%	0.12%	0.10%	0.08%	0.07%

Historical Context

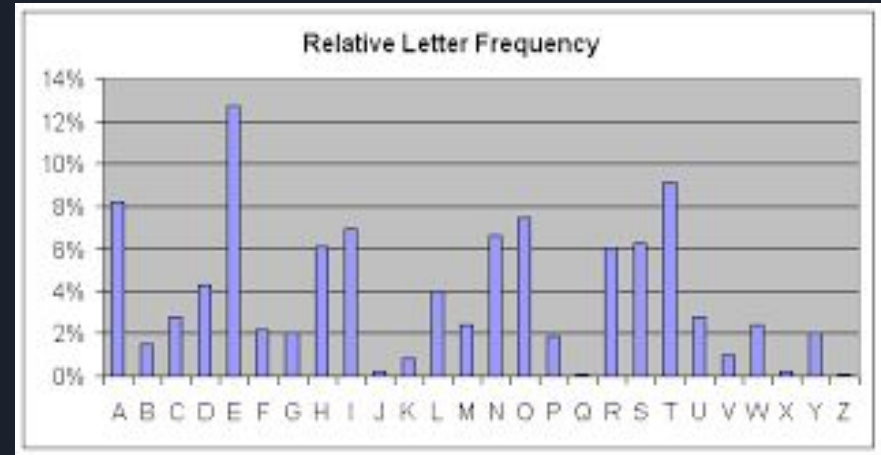
- Arabian mathematician, Al-Kindi, wrote a book that was the first known description of frequency analysis
- Frequency analysis helped during the 1500s and 1600s to break simple substitution ciphers
- More advanced versions were used during World War II to analyze patterns in machine ciphers, which often required some math



A chart shows the frequency of letters in a passage of text.

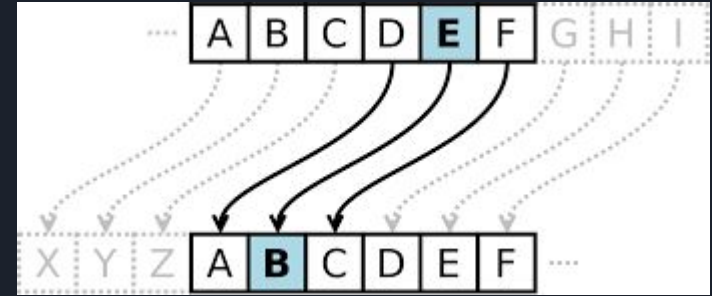
How Codebreakers Use Frequency Analysis

- The codebreaker first counts the occurrences of each letter or symbol in the intercepted ciphertext.
- They then compare these observed frequencies with the known standard letter frequencies of the language
- They make educated guesses about the mappings. For example : The most frequent letter in ciphertext is likely 'E'. The second most frequent is likely 'T', and so on.
- The codebreaker looks for more common words, single-letter words like 'A' or 'I', two letter pairs like 'TH', 'ER' and trigrams like 'THE', 'AND', 'FOR'.
- Through a process of trial and error, the codebreaker refines the key until the message becomes readable.



Example

- Example Using The Caesar Cipher:
 - Encrypted text:
 - jpwolyz hyl jvvs = ciphers are cool
 - Frequencies:
 - j, l, y, v, appear twice
 - p, w, o, z, h, s appear once
 - Common letters include:
 - e, i, o, u, a
 - “vv” is a double letter
 - Possibilities are “oo”, “ee”, etc...
 - Words could be “cool”, “meet”, “seen”, etc...
 - This information can be used to determine trends and other words if you know what some letters equal





Sources

- <https://cryptii.com/pipes/caesar-cipher>
- https://www.princeton.edu/~achaney/tmve/wiki100k/docs/Frequency_analysis.html
- <https://www.khanacademy.org/computing/computer-science/cryptography/ciphers/a/substitution-cipher>
- https://en.wikipedia.org/wiki/Frequency_analysis